



Project Scope Document

Project Name: InventraX

Version: 1.0

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Date: June 11, 2025

1. Project Overview

InventraX is a user-friendly, efficient, and scalable Inventory & Asset Management tool designed to help IT teams, retailers, and office managers effectively track, manage, and organize their equipment, stock, and supplies. This software aims to simplify the process of inventory tracking using a clean interface, barcode integration, and smart stock control features.

2. Objectives

- Provide a centralized system for managing inventory and assets.
 - Allow quick identification of stock levels, locations, and asset assignments.
 - Minimize data entry time with barcode scanning and automation.
 - Enable data-driven decisions through insightful reporting and analytics.
 - Offer a clean, modern, and intuitive interface for all user types.
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3. Features and Functionality

Core Modules:

- **Inventory Dashboard**



- Quick stats, recent activity, low stock alerts.
- **Item Management**
 - Add/edit/delete inventory items.
 - Track categories, quantities, serial numbers, warranty dates.
- **Asset Assignment**
 - Assign assets to users, departments, or locations.
 - View historical records and usage logs.
- **Barcode & QR Integration**
 - Scan and generate barcodes/QRs.
 - Support USB barcode scanners and camera scan.
- **Stock Control**
 - Set reorder levels, manage incoming/outgoing stock.
- **Search & Filters**
 - Quick search by ID, name, category, location, etc.
- **Reports & Analytics**
 - Exportable reports in PDF/Excel/CSV formats.
 - Usage statistics, depreciation, and restock trends.
- **User Roles & Permissions**
 - Admin, Manager, Viewer roles with secure logins.
- **Location Tracking**



- Organize by building, room, shelf, etc.

Optional Future Add-ons:

- **Mobile App Integration**
- **Cloud Sync / Backup**
- **Maintenance Tracking**
- **License & Warranty Management**

4. Out of Scope

- E-commerce integration (no online selling features in v1.0).
- Real-time syncing with external warehouse systems (initially).
- Multi-tenant SAAS features (single org per install in v1.0).

5. Target Users

- IT departments tracking hardware and software assets.
- Retailers managing inventory stock and back-office supplies.
- Offices tracking printer ink, paper, office electronics, etc.
- Small warehouse or storage room managers.

6. Technology Stack



- **Frontend:** Python (Tkinter or PyQt5) or Electron.js (web-based)
- **Backend:** SQLite (v1.0) or PostgreSQL (scalable version)
- **Barcode Tools:** `python-barcode`, `pyzbar`, `qrcode`
- **Reporting:** `pandas`, `matplotlib`, `openpyxl`, `reportlab`
- **Packaging:** PyInstaller for desktop build (Windows/macOS)

7. Timeline

Phase	Description	Duration
Phase 1	UI & Database Design	1 week
Phase 2	Core Functionality Implementation	2–3 weeks
Phase 3	Barcode Integration & Reports	1 week
Phase 4	Testing & Bug Fixes	1 week
Phase 5	Final Packaging & Deployment	1 week

8. Success Criteria

- Software runs smoothly on Windows/macOS.
- Users can add, update, and scan items quickly.
- Reports generate correctly in various formats.



- Positive feedback from early testers (ease of use & functionality).
- Data is securely saved and backed up locally or optionally in the cloud.

9. Constraints

- Must run offline for small businesses without constant internet access.
- Should remain lightweight for low-spec PCs.

10. Ownership & Licensing

This project is created, owned, and maintained by
Alejandro X. Solis
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